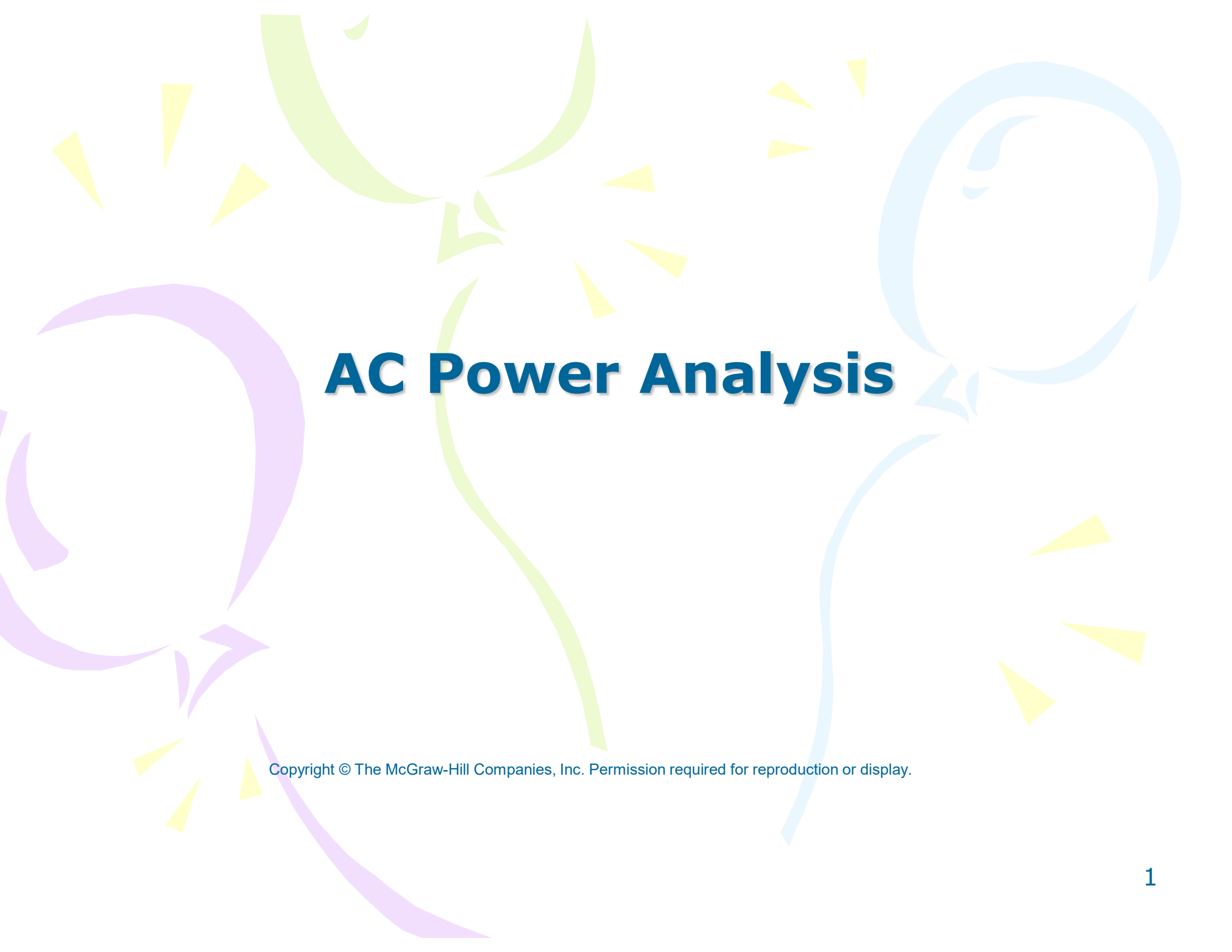


The background features several large, stylized, hand-drawn style swirls in light green, light blue, and light purple. Interspersed among these swirls are numerous small, yellow, four-pointed starburst shapes, some of which are larger and more prominent than others. The overall aesthetic is clean and modern.

AC Power Analysis

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11.1 Instantaneous and Average Power (1)

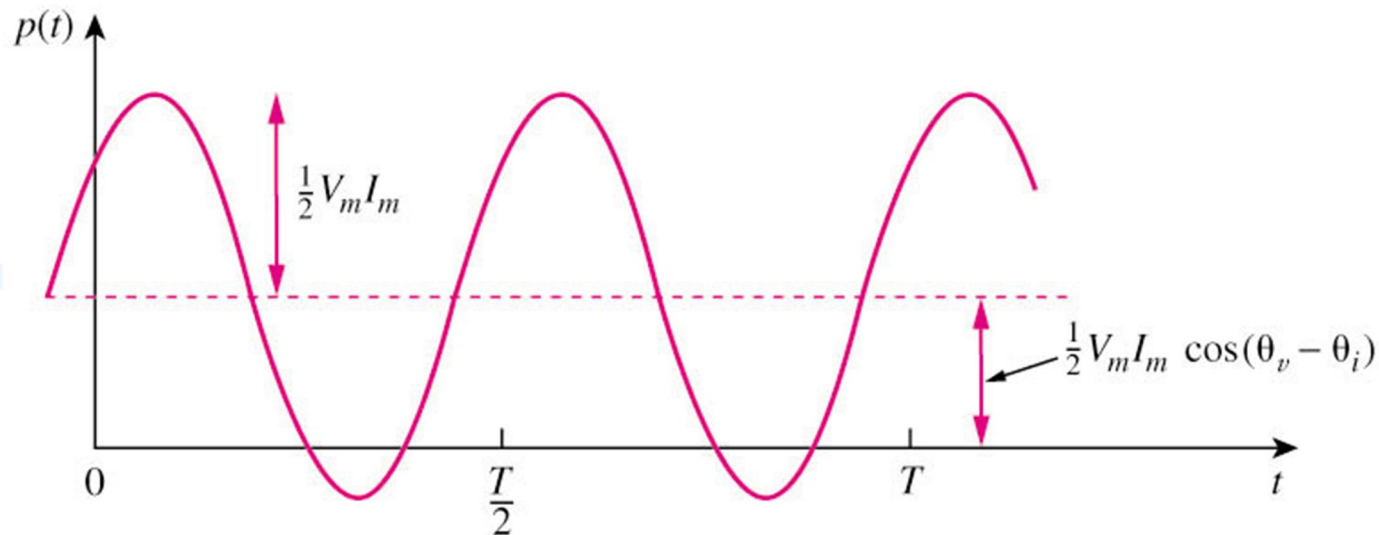
- The instantaneous power, $p(t)$

$$p(t) = v(t) i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Constant power

Sinusoidal power at $2\omega t$

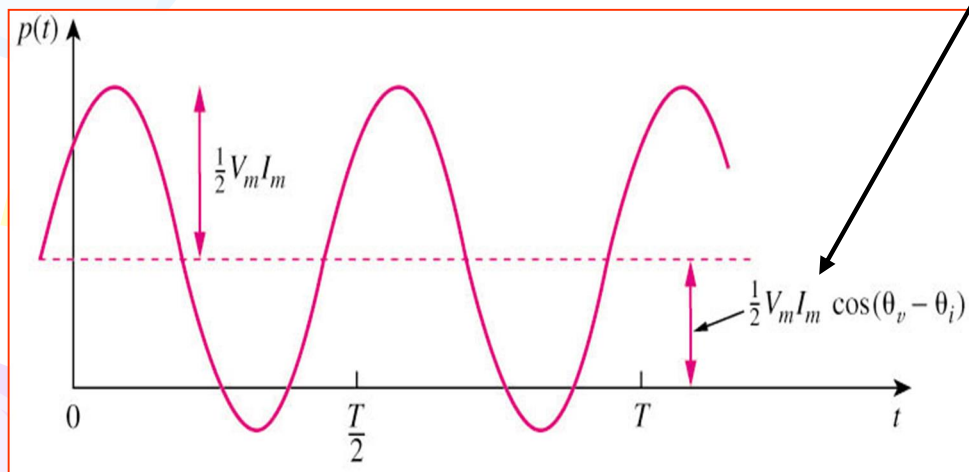


$p(t) > 0$: power is absorbed by the circuit; $p(t) < 0$: power is absorbed by the source.

11.1 Instantaneous and Average Power (2)

- The average power, P , is the average of the instantaneous power over one period.

$$P = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$



- P is not time dependent.
- When $\theta_v = \theta_i$, it is a purely resistive load case.
- When $\theta_v - \theta_i = \pm 90^\circ$, it is a purely reactive load case.
- $P = 0$ means that the circuit absorbs no average power.

11.1 Instantaneous and Average Power (3)

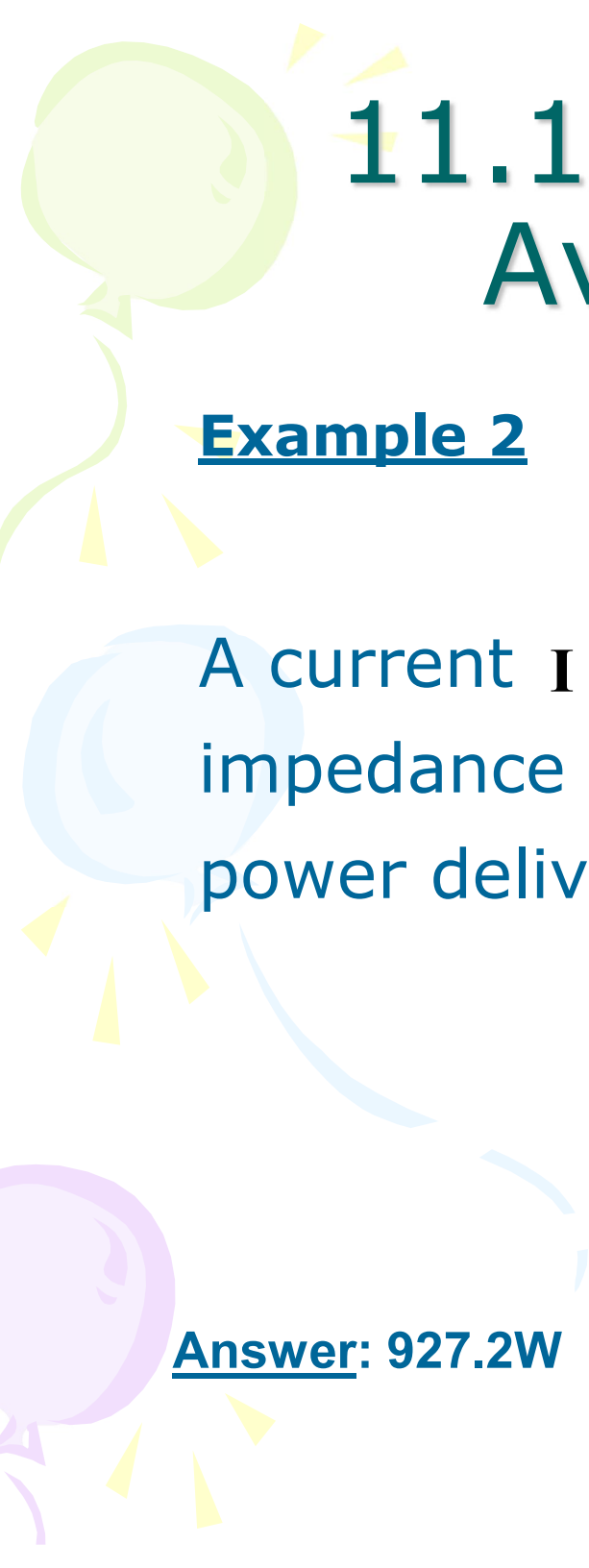
Example 1

Calculate the instantaneous power and average power absorbed by a passive linear network if:

$$v(t) = 80 \cos (10 t + 20^\circ)$$

$$i(t) = 15 \sin (10 t + 60^\circ)$$

Answer: $385.7 + 600 \cos(20t - 10^\circ) \text{ W}, 387.5 \text{ W}$



11.1 Instantaneous and Average Power (4)

Example 2

A current $\mathbf{I} = 10 \angle 30^\circ$ flows through an impedance $\mathbf{Z} = 20 \angle -22^\circ \Omega$. Find the average power delivered to the impedance.

Answer: 927.2W

Example 11.3

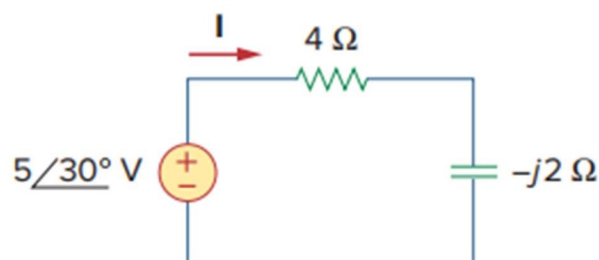


Figure 11.3
For Example 11.3.

For the circuit shown in Fig. 11.3, find the average power supplied by the source and the average power absorbed by the resistor.

Solution:

The current $\mathbf{I}$ is given by

$$\mathbf{I} = \frac{5\angle 30^\circ}{4 - j2} = \frac{5\angle 30^\circ}{4.472\angle -26.57^\circ} = 1.118\angle 56.57^\circ \text{ A}$$

The average power supplied by the voltage source is

$$P = \frac{1}{2}(5)(1.118) \cos(30^\circ - 56.57^\circ) = 2.5 \text{ W}$$

The current through the resistor is

$$\mathbf{I}_R = \mathbf{I} = 1.118\angle 56.57^\circ \text{ A}$$

and the voltage across it is

$$\mathbf{V}_R = 4\mathbf{I}_R = 4.472\angle 56.57^\circ \text{ V}$$

The average power absorbed by the resistor is

$$P = \frac{1}{2}(4.472)(1.118) = 2.5 \text{ W}$$

which is the same as the average power supplied. Zero average power is absorbed by the capacitor.



Practice Problem 11.3

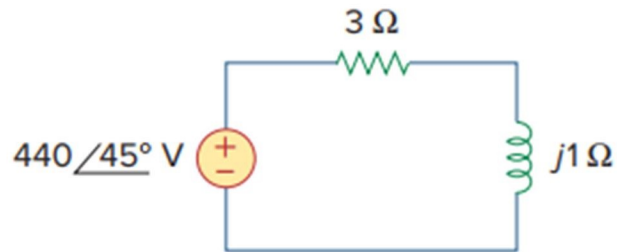


Figure 11.4

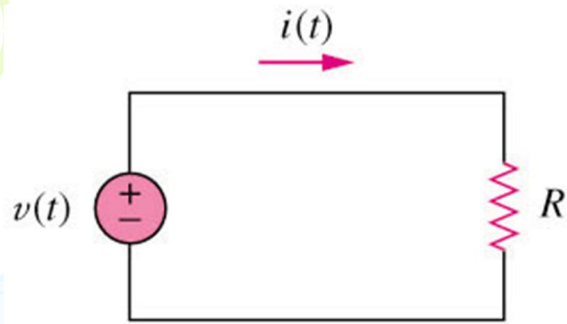
For Practice Prob. 11.3.



In the circuit of Fig. 11.4, calculate the average power absorbed by the resistor and inductor. Find the average power supplied by the voltage source.

Answer: 29.04 kW, 0 W, 29.04 kW.

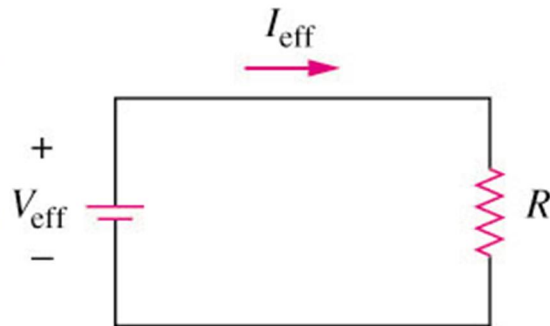
11.3 Effective or RMS Value (1)



(a)

The total power dissipated by R is given by:

$$P = \frac{1}{T} \int_0^T i^2 R dt = \frac{R}{T} \int_0^T i^2 dt = I_{rms}^2 R$$



(b)

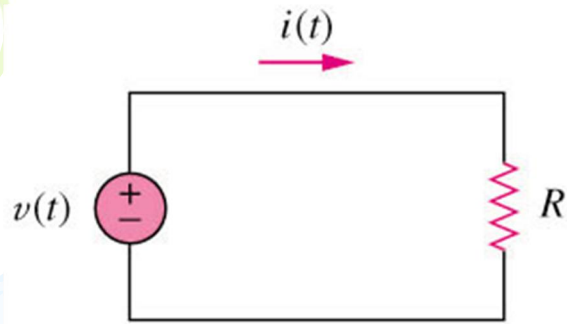
Hence, I_{eff} is equal to:

$$I_{eff} = \sqrt{\frac{1}{T} \int_0^T i^2 dt} = I_{rms}$$

The rms value is a constant itself which depending on the shape of the function $i(t)$.

The effective of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.

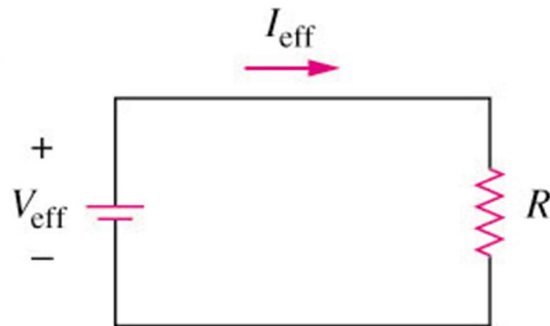
11.3 Effective or RMS Value (2)



(a)

The rms value of a sinusoid $i(t) = I_m \cos(\omega t)$ is given by:

$$I_{\text{rms}}^2 = \frac{I_m^2}{2}$$



(b)

The average power can be written in terms of the rms values:

$$P_{\text{eff}} = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Note: If you express amplitude of a phasor source(s) in rms, then all the answer as a result of this phasor source(s) must also be in rms value.

11.4 Apparent Power and Power Factor (1)

- Apparent Power, S , is the product of the r.m.s. values of voltage and current.
- It is measured in volt-amperes or VA to distinguish it from the average or real power which is measured in watts.

$$P = V_{\text{rms}} I_{\text{rms}} \cos (\theta_v - \theta_i) = S \cos (\theta_v - \theta_i)$$

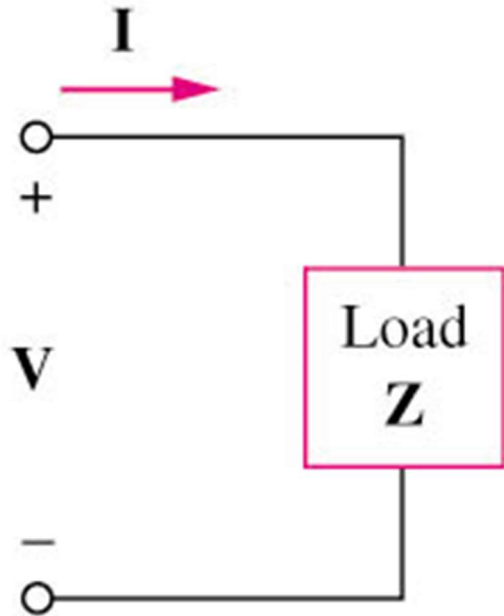
Apparent Power, S

Power Factor, pf

- Power factor is the cosine of the phase difference between the voltage and current. It is also the cosine of the angle of the load impedance.

11.5 Complex Power (1)

Complex power **S** is the product of the voltage and the complex conjugate of the current:

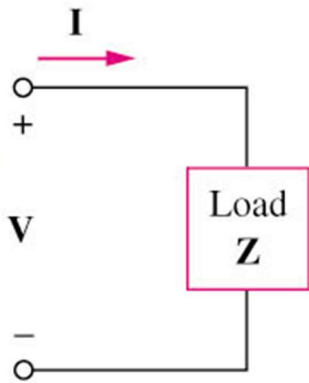


$$\mathbf{V} = V_m \angle \theta_v$$

$$\mathbf{I} = I_m \angle \theta_i$$

$$\frac{1}{2} \mathbf{V} \mathbf{I}^* = V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i$$

11.5 Complex Power (2)



$$S = \frac{1}{2} V I^* = V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i$$

$$\Rightarrow S = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + j V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$$

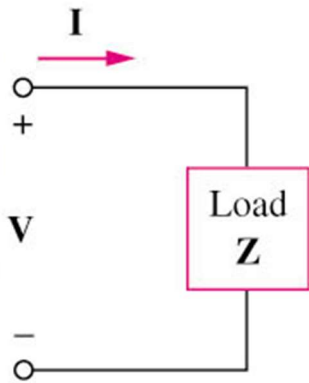
$$S = P + jQ$$

P: is the average power in watts delivered to a load and it is the only useful power.

Q: is the reactive power exchange between the source and the reactive part of the load. It is measured in VAR.

- $Q = 0$ for *resistive loads* (unity pf).
- $Q < 0$ for *capacitive loads* (leading pf).
- $Q > 0$ for *inductive loads* (lagging pf).

11.5 Complex Power (3)



$$\Rightarrow S = \underbrace{V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)}_{\mathbf{P}} + j \underbrace{V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)}_{\mathbf{Q}}$$
$$\mathbf{S} = \mathbf{P} + j \mathbf{Q}$$

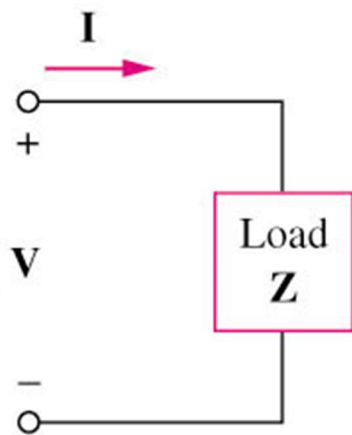
Apparent Power, $S = |\mathbf{S}| = V_{\text{rms}} I_{\text{rms}} = \sqrt{P^2 + Q^2}$

Real power, $P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$

Reactive Power, $Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$

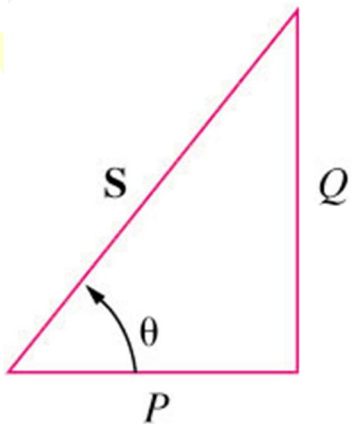
Power factor, $\text{pf} = P/S = \cos(\theta_v - \theta_i)$

11.5 Complex Power (4)

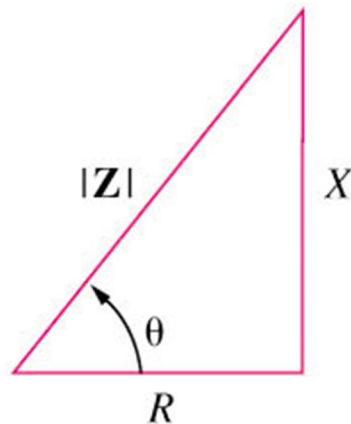


$$\Rightarrow S = \underbrace{V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)}_{\mathbf{P}} + j \underbrace{V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)}_{\mathbf{Q}}$$

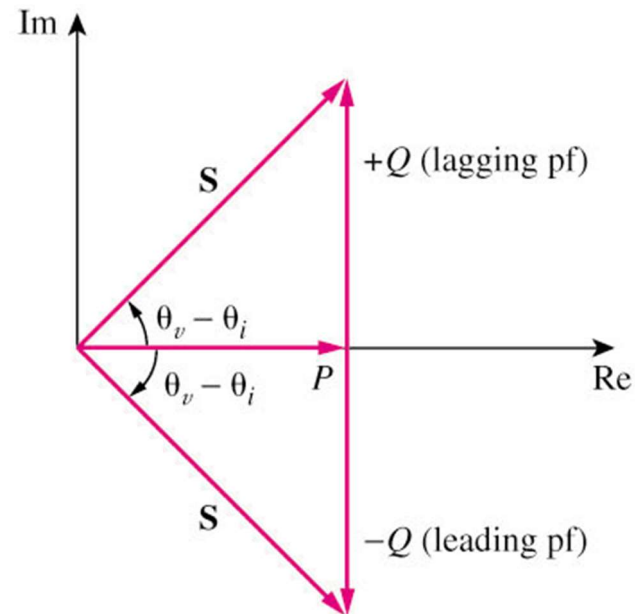
$$\mathbf{S} = \mathbf{P} + j \mathbf{Q}$$



Power Triangle



Impedance Triangle



Power Factor

The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ)$ V and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ)$ A. Find: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Solution:

(a) For the rms values of the voltage and current, we write

$$\mathbf{V}_{\text{rms}} = \frac{60}{\sqrt{2}} \angle -10^\circ, \quad \mathbf{I}_{\text{rms}} = \frac{1.5}{\sqrt{2}} \angle +50^\circ$$

The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = \left(\frac{60}{\sqrt{2}} \angle -10^\circ \right) \left(\frac{1.5}{\sqrt{2}} \angle -50^\circ \right) = 45 \angle -60^\circ \text{ VA}$$

The apparent power is

$$S = |\mathbf{S}| = 45 \text{ VA}$$

(b) We can express the complex power in rectangular form as

$$\mathbf{S} = 45 \angle -60^\circ = 45[\cos(-60^\circ) + j \sin(-60^\circ)] = 22.5 - j38.97$$

Since $\mathbf{S} = P + jQ$, the real power is

$$P = 22.5 \text{ W}$$

while the reactive power is

$$Q = -38.97 \text{ VAR}$$

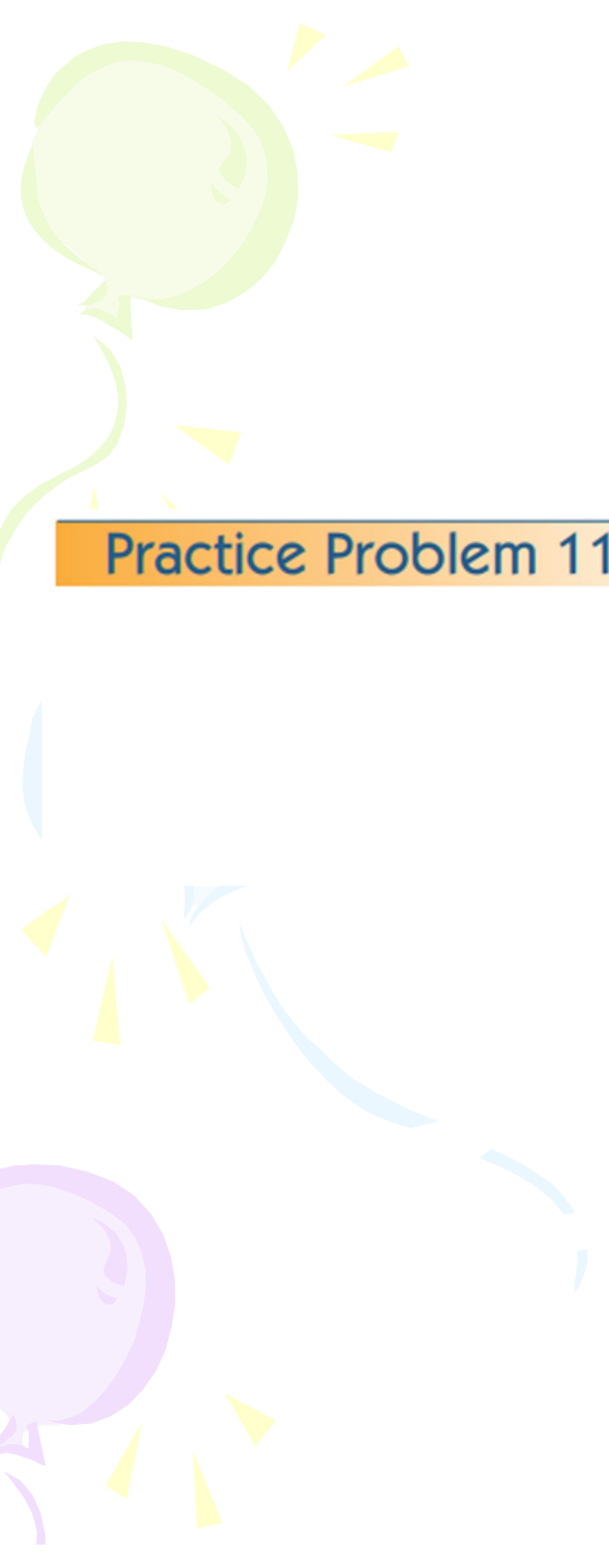
(c) The power factor is

$$\text{pf} = \cos(-60^\circ) = 0.5 \text{ (leading)}$$

It is leading, because the reactive power is negative. The load impedance is

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{60 \angle -10^\circ}{1.5 \angle +50^\circ} = 40 \angle -60^\circ \Omega$$

which is a capacitive impedance.

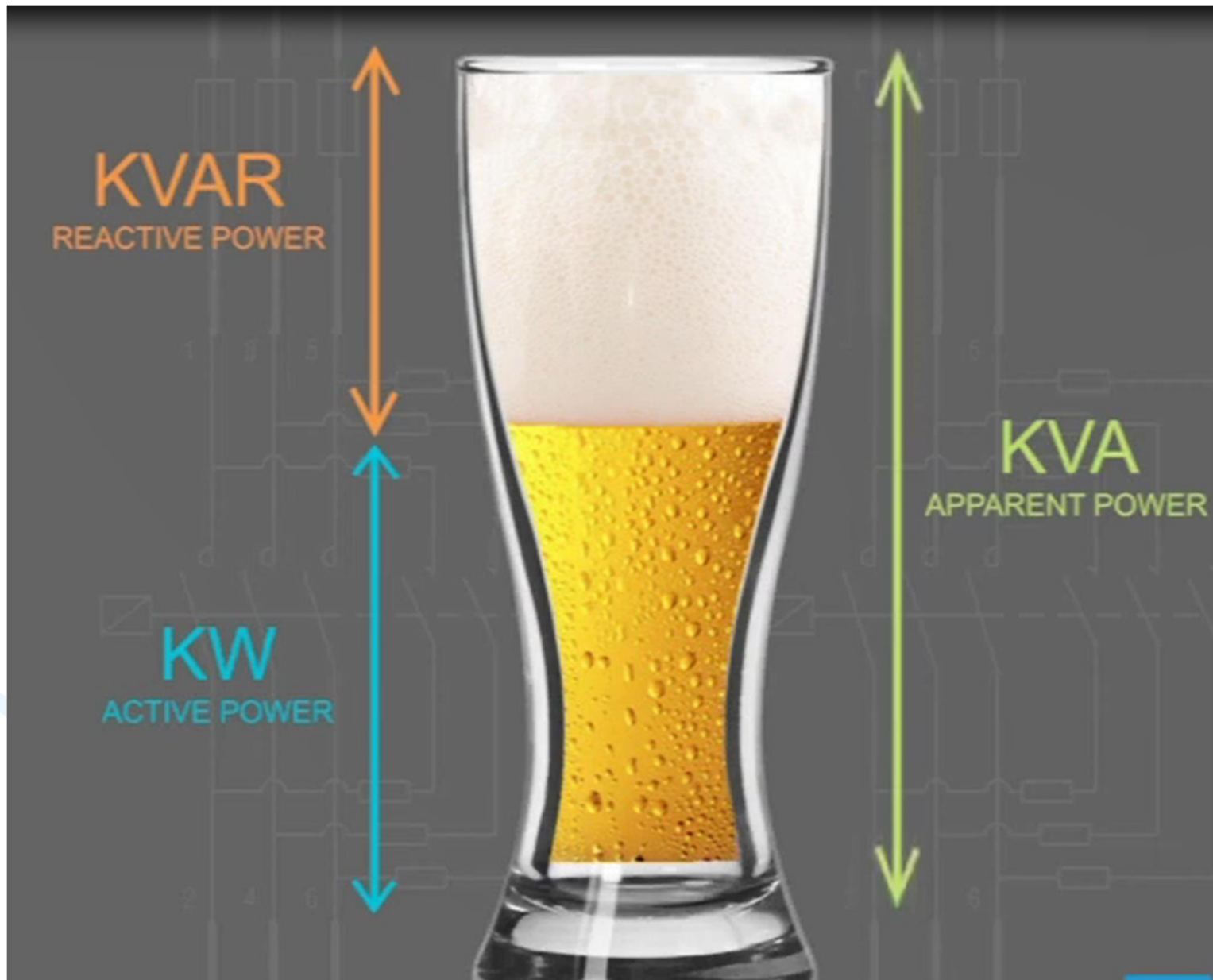


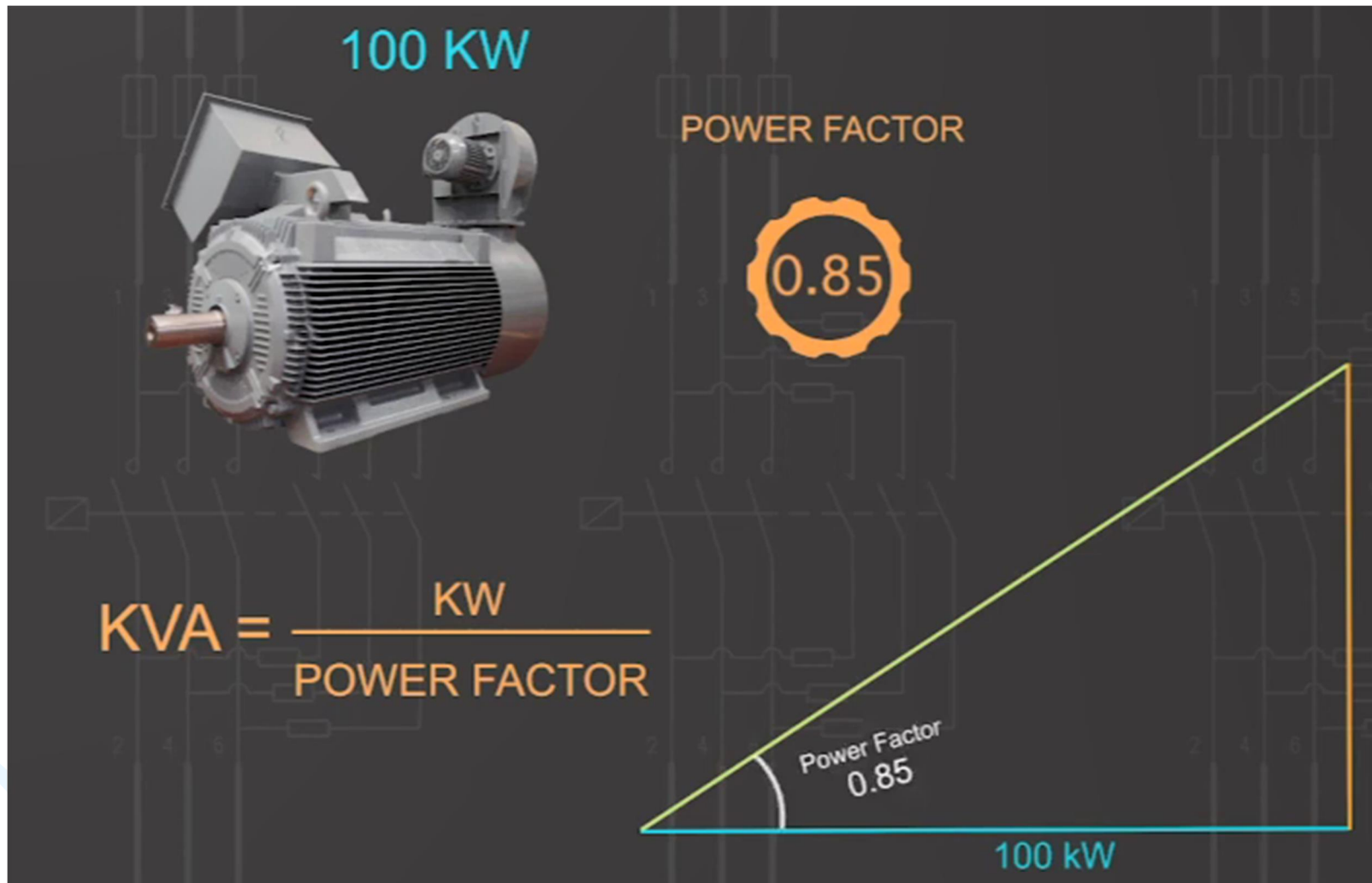
Practice Problem 11.11

For a load, $\mathbf{V}_{\text{rms}} = 110 \angle 85^\circ \text{ V}$, $\mathbf{I}_{\text{rms}} = 3 \angle 15^\circ \text{ A}$. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Answer: (a) $330 \angle 70^\circ \text{ VA}$, 44 VA , (b) 112.87 W , 310.1 VAR , (c) 0.342 lagging, $(12.541 + j34.46) \Omega$.

Power Factor





$$118 \text{ KVA} = \frac{100 \text{ KW}}{0.85}$$



ADVANTAGES

Of Power Factor Correction

Immediate
and
significant
savings on
your
electricity

Improved
plant
efficiency
and longer
equipment
life

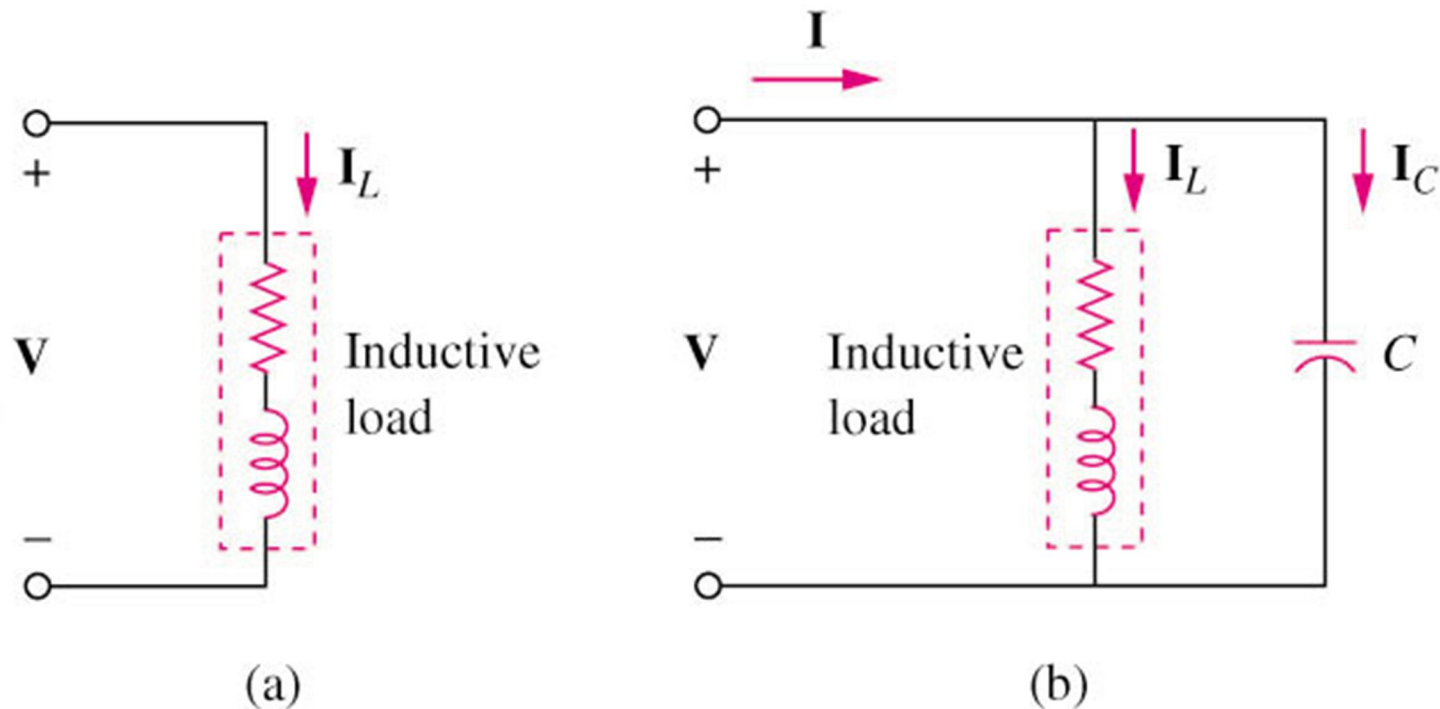
Increased
capacity of
electrical
infrastructure
and
equipment

Elimination
of
low power
factor
penalties

Contribution
towards
reduction in
carbon
footprint

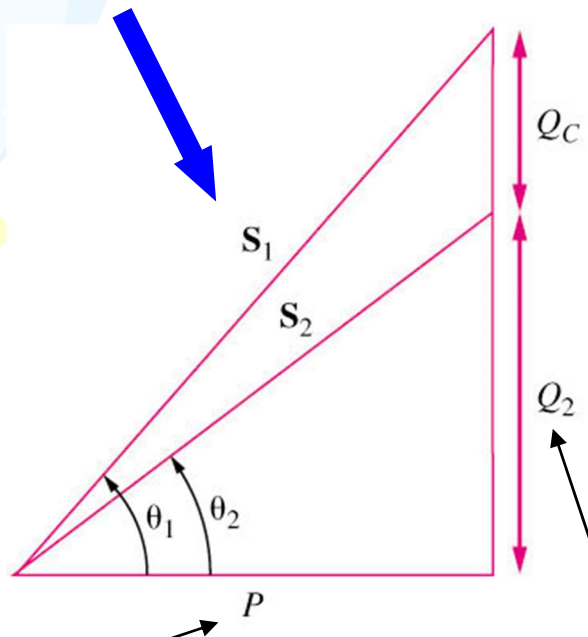
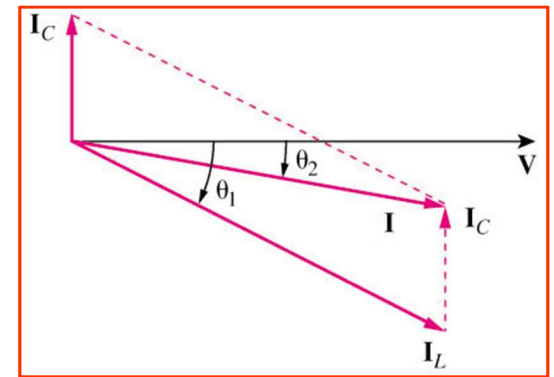
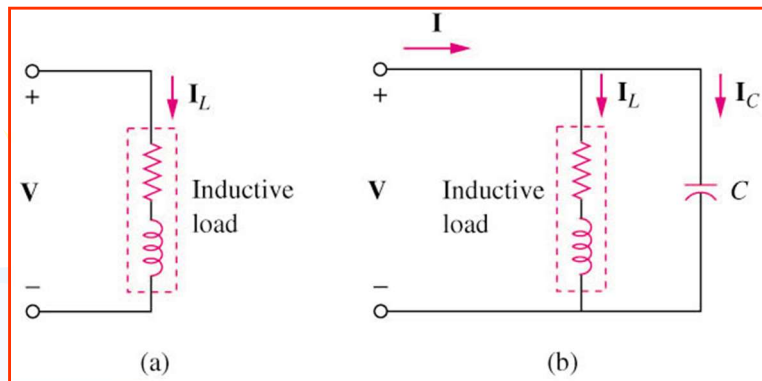
11.7 Power Factor Correction (1)

Power factor correction is the process of increasing the power factor without altering the voltage or current to the original load.



Power factor correction is necessary for economic reason.

11.7 Power Factor Correction (2)



$$Q_c = Q_1 - Q_2$$

$$= P (\tan \theta_1 - \tan \theta_2)$$

$$= \omega C V_{2rms}$$

$$Q_1 = S_1 \sin \theta_1$$

$$= P \tan \theta_1$$

$$P = S_1 \cos \theta_1$$

$$Q_2 = P \tan \theta_2$$

$$C = \frac{Q_c}{\omega V_{rms}^2} = \frac{P (\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

➤ **Practice Problem 11.15:** Find the value of the capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. The load is supplied by a 110 Volt (rms), 60 Hz line.

P.P.11.15 $\text{pf} = 0.85 = \cos \theta \longrightarrow \theta = 31.79^\circ$

$$Q = S \sin \theta \longrightarrow S = \frac{Q}{\sin \theta} = \frac{140}{\sin(31.79^\circ)} = 265.8 \text{ kVA}$$

$$P = S \cos \theta = 225.93 \text{ kW}$$

For $\text{pf} = 1 = \cos \theta_1 \longrightarrow \theta_1 = 0^\circ$

Since P remains the same,

$$P = P_1 = S_1 \cos \theta_1 \longrightarrow S_1 = \frac{P_1}{\cos \theta_1} = 225.93$$

$$Q_1 = S_1 \sin \theta_1 = 0$$

The difference between the new Q_1 and the old Q is Q_c .

$$Q_c = 140 \text{ kVAR} = \omega C V_{\text{rms}}^2 \quad C = \frac{140 \times 10^3}{(2\pi)(60)(110)^2} = \underline{\underline{30.69 \text{ mF}}}$$